

Arthropod diversity based on the prevalence of plants between coffee rows in Costa Rica

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ABSTRACT

Farms have different practices and surrounding conditions which can impact arthropod diversity. Two coffee fields in the Monteverde region were studied under Union Varsan: Turin, which suppresses weeds with roundup to grow beans between the coffee rows, and LIFE, which allows weeds to fully mature between coffee rows. The purpose of this study was to understand how interrow plant prevalence influences arthropod diversity and abundance. Between the coffee rows, five arthropod transects and ten plant quadrants were performed for each field. Arthropods were classified by morphospecies in the laboratory and then True Diversity Index values were calculated for the two fields. LIFE had higher values of mean vegetation cover, mean plant morphospecies, and mean maximum height of plant per quadrant. These higher plant values could be the reason why LIFE had greater arthropod diversity and abundance. The greater diversity of and abundance of arthropods at LIFE can help with functions on the farm such as pollination and reducing pest populations. These findings can demonstrate to farmers the benefits of allowing different plants to grow between coffee rows.

Diversidad de artrópodos según la prevalencia de plantas entre filas de café en Costa Rica

RESUMEN

Las fincas agrícolas practican diferentes prácticas y presentan condiciones circundantes variadas que pueden afectar la diversidad de artrópodos. Se estudiaron dos cafetales en la region de Monteverde que pertenecen a Unión Varsan: Turin, que elimina las malezas con Roundup para cultivar frijol entre las filas de café; y LIFE, donde permiten que las malezas crezcan entre las filas. El propósito de este estudio es comprender cómo la presencia de plantas entre filas de estos dos sitios influye en la diversidad y abundancia de artrópodos. Se realizaron cinco transectos para coleccionar artrópodos y diez cuadrantes para contabilizar las malezas presentes en cada sitio. Los artrópodos se clasificaron por morfoespecie en el laboratorio y luego se calcularon los valores del Índice de Diversidad Verdadera para los sitios. LIFE presentó los valores más altos de cobertura vegetal media, morfoespecies de plantas y altura máxima media de las malezas. Es probable que por este motivo esta finca presentó una mayor diversidad y abundancia de artrópodos. Esta mayor diversidad y abundancia de artrópodos en LIFE podría ayudar con funciones ecosistémicas en la finca como por ejemplo la polinización y la reducción de las poblaciones de plagas. Estos hallazgos pueden demostrar a los finqueros los beneficios de permitir que diferentes plantas crezcan entre las filas de café.

INTRODUCTION

Every farm contains differences with other farms such as practices and surrounding conditions. These differences impact overall arthropod diversity. Arthropoda is the phylum taxonomic order which contains the taxonomic classes of Insecta and Arachnida which were in this study (Lankester, 1904). Some arthropods benefit farmers while others are harmful.

Plants on a farm have an impact on the diversity of arthropods. This project assessed plants between rows- interrow plants - and their impact on arthropod diversity. Intercropping, the practice of planting different crops between the rows of main crops or planting multiple crops mixed within a row, has been shown to increase arthropod diversity in a study taking place in Germany (Brandmeier et al., 2021). Weeds are plants that were not intentionally planted, or are considered undesirable, or are invasive, also play a role in regulating diversity and abundance of arthropods (Norris et al., 2005). Insects may use weeds as a food source or a habitat; some may be completely reliant on specific weeds to complete their life cycle (Marshall et al., 2003).

It is not just the plants within a farm field but also the landscape features that surround it that impact arthropod diversity and abundance. A study in Canada found that closer proximity to forests has been shown to increase arthropod abundance and richness (Mitchell et al., 2014). Distance from rivers can also play an important role. A study in Northern Spain found that greater distance from rivers had a negative effect on aerial arthropods such as wasps as they could be using river resources directly (Albacete et al., 2020). But for ground arthropods, local vegetation composition had a greater impact than the proximity to a river (Albacete et al., 2020). This highlights the importance of plant diversity on farm plots and diversity of surrounding vegetation such as forests on arthropod diversity and abundance.

Due to their small nature, many farmers may be overlooking the impact arthropods have on their farms. There are many arthropods that damage coffee plants and are considered pests as a result. They damage seedlings, reduce coffee yield, and reduce the quality of the beans (Shimales and Alemayehu, 2024). On the other hand, there are many beneficial arthropods, some of which feed upon the coffee pests (Shimales and Alemayehu 2024), while others, in particular bees, play an important role in pollinating coffee plants. In one study located in Costa Rica, coffee yield was reduced by 24.5% with bees excluded, showcasing how pollinators boost coffee yield (Martinez-Salinas et al., 2022).

The farms chosen for this research project work under the Union Varsan, which is made up of 21 families in Monteverde (D. Mbeki, pers. comm. 2025). They strive for sustainability in coffee production, forest conservation, social justice for their workers, and education (E. Solano, personal communication, 2025). One of their farms, Finca LIFE, uses half of its land to produce coffee, while the other half is conserved as forest (E. Solano, pers. comm. 2025). La Paila, a farm plot within Finca LIFE was one of the chosen study sites since they allow weeds to grow between their coffee plant rows. The second farm plot, called El Recibidor, is within Finca Turín and was chosen because they suppress weeds with herbicides in order to grow beans between their coffee plant rows. Both of these farms are in the Monteverde region.

While there is a significant amount of known information on how intercropping, weeds, forests, and rivers affect arthropod diversity, the impact of the unique levels of these variables for Union Varsan's farms is not known. This project performed a targeted study for Union Varsan, finding how differences in interrow plant prevalence influences arthropod diversity and abundance between Turín and LIFE. This could help coffee farmers understand how their interrow plant management impacts arthropods, and how arthropods may benefit or harm coffee farms. Thus this study's question was: Is there a difference in arthropod diversity between the two farms based on interrow plant prevalence?

MATERIALS AND METHODS

Two fields were surveyed to determine the influence of interrow plants on arthropod diversity. Both are in the Monteverde region of Costa Rica. El Recibidor is a field within Finca

Turin. La Paila is a field within Finca LIFE. I chose these two fields because they have different interrow plant prevalence from which I can compare arthropod diversity. Plant prevalence is defined as the vegetation coverage, plant morphospecies richness, and maximum height of plants in this study. Vegetation coverage is the percent of bare ground covered by alive leafy material.

Based on observation, El Recibidor has less interrow plant prevalence than LIFE. El Recibidor applies herbicides up to three times a year to suppress weed growth in between the four month old coffee plant rows. Suppressing the weeds also served to free up space to plant culinary beans between the rows for additional food production. Of its four surrounding sides, El Recibidor has three sides that are one row of windbreak trees then more coffee fields. The fourth side is a one meter weed break followed by another coffee field. There were bananas and other trees interspersed throughout the field. El Recibidor sits at an elevation of 1000 meters above sea level and is 600 meters from the nearest river (E. Álvarez Aleman, pers. comms. 2025). Foliar fungicide was applied to the coffee plants around 60 days before the data collection days (E. Álvarez Aleman, pers. comm. 2025).

La Paila does not use herbicide to suppress weed growth. Instead, they use a machete to cut weeds that are in a one meter radius of the young coffee plants (E. Álvarez Aleman, pers. comm. 2025). Then in a one meter break between the coffee rows, the farmers do not cut the weeds and allow them to grow to maturity, flower, and produce seed. Of its four surrounding sides, La Paila has two sides that are forest. Another side is a forest break then a banana plantation, and the fourth side is a one layer windbreak followed by another coffee field. There were banana and citrus trees interspersed throughout the fields. La Paila sits at an elevation of 1100 meters above sea level and is 200 meters from the nearest river (E. Álvarez Aleman, pers. comm. 2025). An organic fungicide called Trichoderma was applied to the roots of the coffee plant when they were planted (E. Álvarez Aleman, pers. comm. 2025).

Field collection

To collect insects in the field, five transects were done at each field. Insect transects at Turín were done in the morning of November 11, 2025 before 12 pm, and in LIFE they were done in the morning of November 17, 2025. Each transect was between the rows of coffee and 10 meters in length. Insects were collected from one end of the field to the other. This was done in order to minimize arthropod disturbance before transect collection was done. The locations of the transects were chosen in various locations and conditions around the farm in order to account for the variety within each field. Samples included variations in proximity to the edges of the field, levels of sunlight and shade, and vegetation between rows. Once an area was chosen to be a transect, I used an aerial net to sweep from left to right with an aerial net, constantly moving forward at a walking pace. Once I reached the end of the transect, the insects were transferred from an aerial net to a plastic bag. Once I arrived at the lab, I froze the arthropods in a freezer.

I collected ten quadrats of interrow plants at each farm. The ten quadrats at Turín were collected on November 21, 2025. The ten quadrats at LIFE were collected on November 17, 2025. Each farm had pretty homogenous interrow plant conditions. To determine the percentage of vegetation coverage in between the rows of coffee plants, I used a 1 by 1 meter quadrat. I placed quadrats throughout each field with a similar selection process to that of collecting insects. Each quadrat measured 1 meter by 1 meter. To calculate vegetation coverage, the quadrat was divided into four sections. Vegetation coverage in each quarter was estimated to the nearest five percent, and an average of the four quarters resulted in the vegetation coverage estimate. For quarters of a quadrat that only had minimal weed coverage but that had one weed individual present, the percent coverage for that quarter was considered to be 5 percent. After collecting the

vegetation coverage values I collected as many unique plant morphospecies that I could find in order to get the plant morphospecies richness per quadrat. The plant morphospecies were stored in a plastic bag to be identified later in the laboratory. Finally, I measured the height of the tallest plant I could find within the quadrat.

Laboratory work

Laboratory work was done at the Monteverde Institute. Collected insects were frozen in the lab's refrigerator for at least 30 minutes. After the insects were frozen, they were emptied onto blank white paper in order to separate the arthropods from plant debris and soil. Each specimen was placed in 70% ethanol in a vial corresponding to the transect number they were captured in.

Identification began by grouping similar species based on visual distinctions with the naked eye. Then, more specific groupings were done under a dissecting microscope. Insects were identified based on taxonomic order and species morphologies. Species morphologies were based on differences in color, size, and body parts. To create a list for morphospecies, a picture of each morphospecies was taken with written descriptors of a morphospecies color, size, body parts, and traits to remember the morphospecies. Creating a new morphospecies was done on the more conservative side, aiming for more certainty in each morphospecies classified but potentially leading to an underestimation of the true number of species present and grouping of different morphospecies into one morphospecies.

For each transect sample the quantity of each insect morphospecies was recorded. This data was used to calculate True Diversity Index values.

In addition, for each field, I marked down how many new insect morphospecies were classified after each successive transect sample. By recording this, I was able to run a complementary analysis on the cumulative morphospecies collected at each site (Figure 4). The purpose of complementary analysis was to see if I collected all of the insect morphospecies that were present within each field. To determine this I could look at the cumulative species curve and see if it plateaus, or reaches a limit, reaching an estimate of the full arthropod morphospecies present. I added a least-squares regression curve to the five transect points to help visualize the trend of the number of new arthropod morphospecies being added (Figure 4).

As for plant morphospecies, collected weeds for the ten samples from each farm were identified with the help of local naturalist Eladio Cruz. The number of unique plant morphospecies per sample was recorded, and all were classified to the family taxonomic level with some up to the species level. The identification process did not classify one species as two species for all of the collected plants.

Data analysis

Data analysis was performed using the application R Studio (version 4.4.1). To compare the average vegetation coverage, average number of plant morphospecies, and average maximum plant height between the two farms, a two sided t test was performed for the three variables.

To compare diversities between the two farms the True Diversity Index (Hill Numbers) was used. The True Diversity Index equation has a q value that can be changed in order to obtain different species calculations. When $q = 0$, all species are weighted equally. This produces a raw species richness value. When $q = 1$, species are weighted proportional to the number of individuals they have. This produces a calculated number of species if they were all balanced. When $q = 2$, more abundant species are weighted more heavily. This produces a calculated number of dominant species.

RESULTS

Overall, analysis of plant data found that LIFE had higher interrow plant prevalence. LIFE had greater vegetation cover ($p < 0.01$, Figure 1). LIFE also had greater plant morphospecies ($p < 0.01$, Figure 2). LIFE also had greater maximum heights of plants between the coffee rows ($p < 0.01$, Figure 3). Overall, LIFE, which uses organic manual weeding practices, had greater interrow plant prevalence compared to Turín, which uses herbicide to control interrow plants.

Figure 1 displays that LIFE had higher vegetation cover compared to Turín. This bar plot shows this difference by displaying the average vegetation cover at the two farm sites. The lines on each bar are error bars of one standard deviation above and below the average. I found a statistically significant difference in average vegetation coverage between the two fields. ($T = 23.261$, $df = 16.536$, $p\text{-value} = 4.564 \times 10^{-14}$).

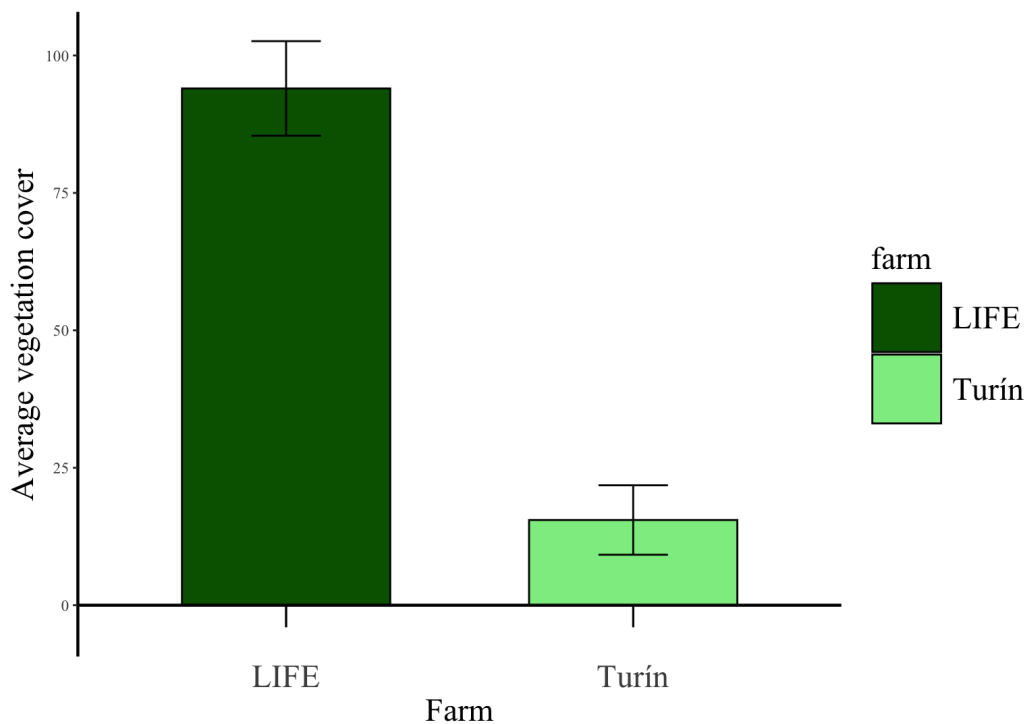


Figure 1: Average vegetation coverage between coffee rows at each farm

Figure 2 demonstrates that LIFE had greater average values of plant morphospecies compared to Turín. This bar plot shows this difference by displaying the average plant morphospecies at the two farm sites. The lines on each bar are error bars of one standard deviation above and below the average. I found a statistically significant difference in average plant morphospecies between the two fields ($t = 10.002$, $df = 13.056$, $p\text{-value} = 1.729 \times 10^{-7}$).

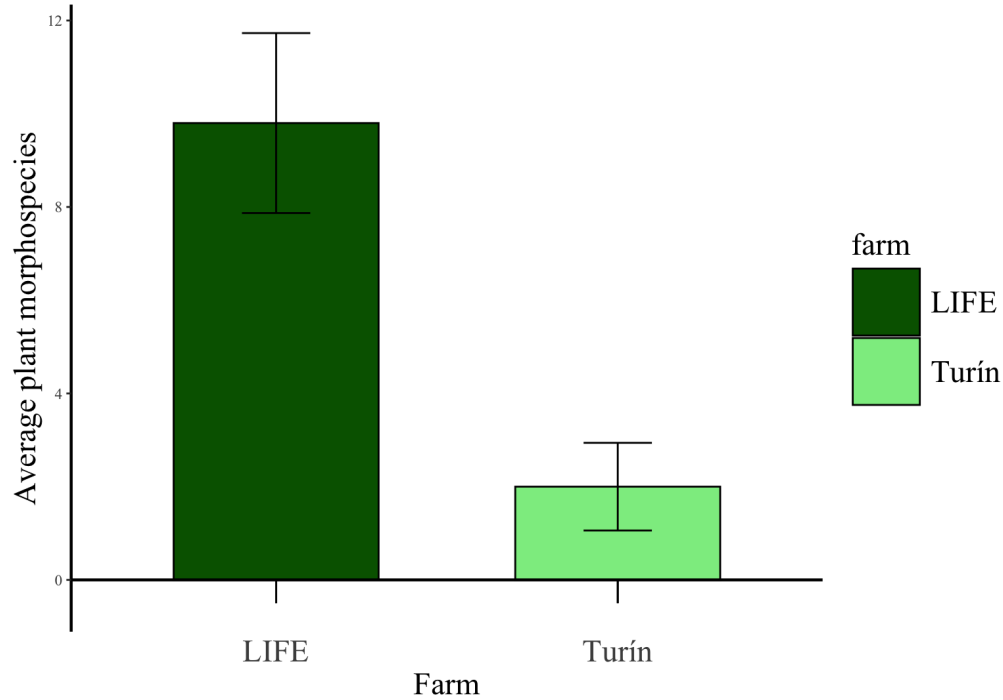


Figure 2: Average plant morphospecies between coffee rows at each farm

Figure 3 demonstrates that LIFE had greater average plant heights between coffee rows compared to Turín. The greater amount of vertical space could provide more habitat for insects (source). This bar plot shows the difference by displaying the average maximum plant height in centimeters at the two farm sites. The lines on each bar are error bars of one standard deviation above and below the average. I found a statistically significant difference in average height of the tallest weed between the two fields ($t = 8.555$, $df = 9.7383$, $p\text{-value} = 7.756 \times 10^{-7}$).

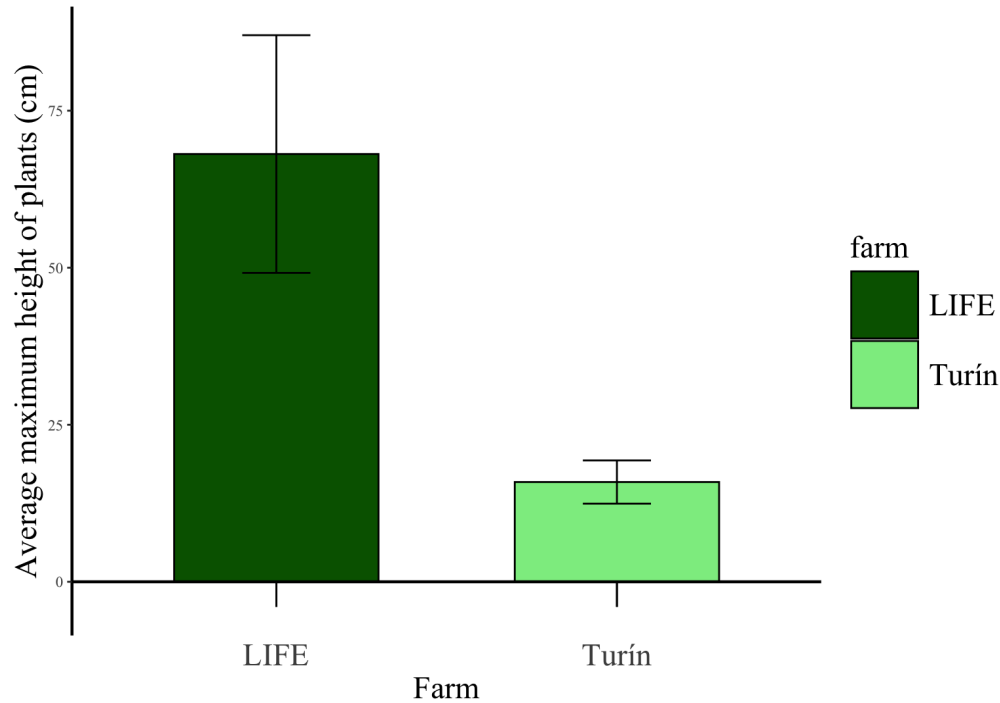


Figure 3: Average maximum height of plants between coffee rows at each farm

Not all the arthropod morphospecies actually present within each field were collected. A complementary analysis found that five transects at each field were insufficient to estimate the true and full amount of arthropod morphospecies at each field (Figure 4). After five transects, LIFE reached a greater arthropod morphospecies of 98 compared to Turin, which reached 44 arthropod morphospecies (Figure 4).

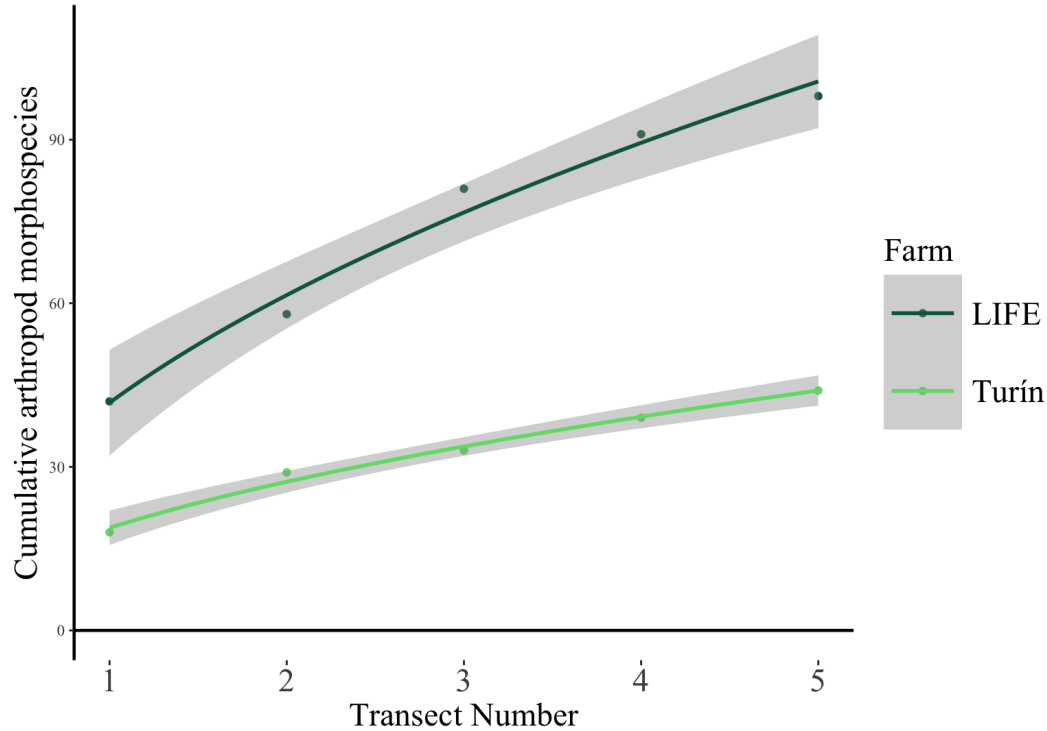


Figure 4: Complementary analysis curve of Turin and LIFE's arthropod morphospecies

Arthropod diversity results

LIFE had greater arthropod diversity than Turin. This is demonstrated by the True Diversity Index that showed that LIFE had greater values than Turin in species richness ($q = 0$), calculated number of even species ($q = 1$), and calculated number of dominant species ($q = 2$) (Table 1).

Table 1: True diversity index values across the two farms

	Turin	LIFE
Species richness ($q = 0$)	44	98
Balanced species ($q = 1$)	19.97	34.33
Dominant species ($q = 2$)	11.26	19.20

Arthropod abundance results

LIFE had a greater adult arthropod abundance than Turin. Totaling the adult arthropod abundance for the five transects at each site found that LIFE had 679 arthropod adults, compared to Turin which had 173. An interesting finding was that all the 7 Lepidoptera collected were all from LIFE. Another interesting finding was that 53 spiders were collected in LIFE compared to Turin, where only 1 spider was collected.

DISCUSSION

LIFE had greater arthropod diversity and abundance than Turin. This could have been because of the greater interrow plant prevalence in between its coffee rows. Finding greater interrow weeds at LIFE is consistent with a meta - analysis of farms in Europe which found that organic methods of controlling weeds lead to greater weed density and weed diversity compared to conventional farming practices which utilize herbicides (Mwangi et al, 2024). The greater interrow plants provide more food resources and structures to live within (Marshall et al., 2003). Collecting all the Lepidoptera at LIFE also shows how the interrow plants might be supporting greater diversity as butterflies require specific hosts to complete their lifecycle (Janz and Nylin, 2017). Another factor that could have increased arthropod diversity at LIFE could have been that it had more forest surrounding its field compared to Turin which mostly had one layer windbreaks then coffee fields around it (Mitchell et al., 2014). Furthermore, LIFE's closer to proximity to a river could have increased its arthropod diversity (Albacete et al., 2020). Nevertheless, the Rakotomalala and coworkers' meta analysis across 18 countries supports that interrow plants remain a significant factor to arthropod diversity and abundance (2023).

A potential source of error in determining arthropod diversity at each site was the morphospecies classification. Morphospecies classification was made on the more conservative side with certain orders such as Hemiptera but especially with Diptera insects, meaning that morphospecies was underestimated. Another source of error could have been the classifications of new insect morphospecies becoming more specific as the later transects were identified and I became more adept at classification. The transects that were from LIFE were identified after the five transects from Turin. This could mean that if identification became more specific over time then the morphospecies at LIFE could have been overestimated. Nevertheless, there was a constant effort to keep the specificity of classifying morphospecies similar to the level it was for

the earlier five transects therefore this was likely not too significant. Furthermore, the cumulative analysis showed that the full diversity of arthropod morphospecies at each site was not collected. In order to fully survey the full diversity at each site, future studies could perform more transects.

While more interrow plant quadrats could be collected at each field in future studies to strengthen confidence in the findings, each field was pretty homogenous in its interrow plant composition. Therefore ten quadrats was likely sufficient in comparing the two fields.

LIFE's practices which increase arthropod diversity and abundance can provide many beneficial ecosystem services in return. Greater presence of arthropod pollinators, from LIFE's increased interrow plants that provide more attractive flowers can help foster more coffee pollination as well boosting coffee yields (Martinez-Salinas et al., 2022). In addition, greater arthropod diversity and abundance has been shown to reduce pest pressure (Rakotomalala et al., 2023). This reduced pest pressure can prevent yield loss and control disease spread between coffee plants. The reduced pest pressure is a result of increased predatory arthropods as well as pests feeding on the interrow weeds instead of the coffee plants (Rakotomalala et al., 2023), (Sharma et al., 2018). The result of all the spiders but one being found in LIFE can serve to highlight the increased pest control services at LIFE (Maloney et al., 2003). These services from arthropods demonstrate to farmers the benefits of allowing more plants to grow between coffee rows.

So what stops a farm like Turin from adopting the practices of LIFE? For one, the manual weeding practices at LIFE must be performed more often than herbicide weed control and requires more labor overall (E. Álvarez Aleman, pers. comm. 2025). This requires more effort on the part of farmers and an increased financial cost of around 200 U.S. dollars per 1 hectare field per year compared to applying herbicide (E. Álvarez Aleman, pers. comm. 2025). And while there is benefit to allowing weeds to grow tall and to maturity, interrow plants must still be cut before harvest because snakes can hide in the grass and bite the workers (E. Álvarez Aleman, pers. comm. 2025). This is an important balance that must be struck between fostering arthropod diversity and abundance with mature interrow plants versus maintaining social sustainability by protecting coffee harvesters.

While there are barriers to implementing practices that promote interrow plant prevalence on coffee farms, there are ways to help farmers overcome them. Farmers can be paid the difference to implement manual weeding practices that are often more costly and labor intensive. Funding for this could be sourced from charging premiums to consumers who are willing to pay for more sustainable produced products, government subsidies, or non - profits who assist in regenerative agricultural implementation. The payments can be used to hire more workers or pay existing workers which serves the side benefit of providing more income to farm workers. In terms of educational barriers, LIFE is currently working on providing educational workshops to other coffee farmers on how they can implement practices such as fostering interrow plants.

By working towards emulating practices of LIFE, farmers can help benefit arthropod diversity and abundance, which can help farmers in return.

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